

No overcomplete lattice Gabor frame is a bi-infinite operator orbit

A full answer to the Gabor part of the question in arXiv:1804.03438

Autonomous solve-first investigation, lane 12

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Abstract

Christensen, Hasannasab, and Philipp ask whether an overcomplete Gabor frame can, after some ordering by $\mathbb{Z}$, be represented as a full orbit $(T^n f_0)_{n \in \mathbb{Z}}$ of one bounded operator. Corso later proved nonexistence for irrational lattice density and for one compact-support class, leaving general rational density open. We give a negative answer in full: no overcomplete one-window lattice Gabor frame has such an ordering. After canonical tightening, a hypothetical orbit becomes unitary. Its threshold-correlation graphs are Cayley graphs of $\mathbb{Z}$ and have linear growth, whereas a nondegenerate Gabor correlation graph contains a quadratically growing $\mathbb{Z}^2$ subgraph. In the remaining rank-one case, the two graphs have different numbers of connected components.

The question and its later refinement. For $a, b > 0$ and $g \in L^2(\mathbb{R}) \setminus \{0\}$, write

$$\pi(m, n) = M_{mb} T_{na}, \quad \mathcal{G}(g, a, b) = \{\pi(m, n)g : (m, n) \in \mathbb{Z}^2\}.$$

The source paper asks whether there exist overcomplete Gabor frames that, under an ordering by $\mathbb{Z}$, are $(T^n f_0)_{n \in \mathbb{Z}}$ for a bounded operator T [1, p. 4]. Negative powers and the frame property force T to be boundedly invertible. Corso subsequently ruled this out when ab is irrational and when g has support in an interval of length a , but explicitly left the remaining rational-density case open [2, pp. 10–11].

Theorem 1 (full Gabor answer). *Let $\mathcal{G}(g, a, b)$ be an overcomplete frame for $L^2(\mathbb{R})$. There are no bijection $\varphi : \mathbb{Z} \rightarrow \mathbb{Z}^2$, vector f_0 , and $T \in GL(L^2(\mathbb{R}))$ such that*

$$\pi(\varphi(n))g = T^n f_0 \quad (n \in \mathbb{Z}).$$

Thus the answer to the Gabor part of the source question is negative for all lattice densities and all nonzero L^2 windows.

1 Two reductions

Lemma 2 (canonical tight orbit). *If the ordering in Theorem 1 existed and S were the Gabor frame operator, then, with $\gamma = S^{-1/2}g$,*

$$t_n := \pi(\varphi(n))\gamma = U^n t_0, \quad U = S^{-1/2} T S^{1/2}, \quad (1)$$

would be an overcomplete Parseval frame generated by a unitary U .

Proof. The Gabor frame operator commutes with every lattice shift $\pi(m, n)$: conjugating the defining frame-operator sum by one shift merely reindexes its rank-one terms (the projective phases cancel). Functional calculus therefore gives $S^{-1/2}\pi(m, n) = \pi(m, n)S^{-1/2}$, proving (1). The canonical tight frame is Parseval and remains overcomplete because $S^{-1/2}$ is invertible.

For rank-one operators written $x \otimes x$, strong convergence of the Parseval frame-operator sum gives

$$UU^* = U\left(\sum_{n \in \mathbb{Z}} t_n \otimes t_n\right)U^* = \sum_{n \in \mathbb{Z}} t_{n+1} \otimes t_{n+1} = I.$$

Since U is invertible, it is unitary. □

Put

$$r_k = \langle t_0, U^k t_0 \rangle \quad (k \in \mathbb{Z}).$$

Lemma 3 (decay of unitary-orbit correlations). *For the Bessel unitary orbit in Lemma 2, $r_k \rightarrow 0$ as $|k| \rightarrow \infty$. Hence, for every $\varepsilon > 0$,*

$$Q_\varepsilon = \{k \in \mathbb{Z} \setminus \{0\} : |r_k| \geq \varepsilon\} \quad (2)$$

is finite.

Proof. Let μ be the scalar spectral measure of U at t_0 . If $p(z) = \sum c_k z^k$ is a trigonometric polynomial, the Bessel synthesis bound gives

$$\int_{\mathbb{T}} |p|^2 d\mu = \|p(U)t_0\|^2 \leq B \sum_k |c_k|^2 = B \int_{\mathbb{T}} |p|^2 dm.$$

Thus $d\mu = w dm$ for some $w \in L^\infty(\mathbb{T})$. Since $r_k = \int_{\mathbb{T}} z^k w(z) dm(z)$, the Riemann–Lebesgue lemma proves the claim. □

Time-frequency covariance gives, for $v, \lambda \in \mathbb{Z}^2$,

$$|\langle \pi(\lambda)\gamma, \pi(\lambda + v)\gamma \rangle| = |a(v)|, \quad a(v) := \langle \gamma, \pi(v)\gamma \rangle. \quad (3)$$

Only the magnitude is used, so the Weyl cocycle causes no issue. Define

$$D = \{v \in \mathbb{Z}^2 \setminus \{0\} : a(v) \neq 0\}, \quad H = \langle D \rangle \leq \mathbb{Z}^2. \quad (4)$$

The bijection φ in (1) preserves all inner-product magnitudes:

$$|a(\varphi(j) - \varphi(i))| = |r_{j-i}|. \quad (5)$$

2 Correlation-graph obstruction

Regard a family of vectors as a graph by joining two distinct indices exactly when their inner product is nonzero. By (3), the Gabor graph has components equal to the cosets of H . By (5), the unitary-orbit graph has components equal to the cosets in $\mathbb{Z}$ of

$$K = \langle Q \rangle, \quad Q = \{k \in \mathbb{Z} \setminus \{0\} : r_k \neq 0\}. \quad (6)$$

Moreover, φ is an isomorphism between these graphs.

Suppose first that $\text{rank } H \leq 1$. Then H has infinite index in $\mathbb{Z}^2$, so the Gabor graph has infinitely many connected components. On the other hand, $Q \neq \emptyset$: if all off-diagonal r_k vanished, the Parseval orbit would be an orthonormal basis, contrary to overcompleteness. Hence $K = d\mathbb{Z}$ for some $d \geq 1$, and the orbit graph has exactly d components. This contradicts graph isomorphism.

It remains to treat $\text{rank } H = 2$. Choose linearly independent $v, w \in D$ and fix

$$0 < \varepsilon < \min\{|a(v)|, |a(w)|\}.$$

Join indices when their inner-product magnitude is at least ε . On $\mathbb{Z}^2$ this threshold graph contains the edges with steps $\pm v, \pm w$. Thus its radius- R ball about any vertex contains

$$\{mv + nw : |m| + |n| \leq R\},$$

which has $1 + 2R(R + 1)$ elements because v, w are independent.

Under φ , this is isomorphic to the threshold graph on $\mathbb{Z}$ with step set Q_ε from (2). Let $M = \max\{|q| : q \in Q_\varepsilon\}$; the set is nonempty because the v -edge has an image. Every radius- R ball in this graph lies in an integer interval of length $2MR$, so it has at most $2MR + 1$ vertices. For large R this contradicts the quadratic lower bound. Both possible ranks of H are impossible, completing the proof of Theorem 1. $\square$

Scope. The proof closes the Gabor existence question, including the rational-density case left in [2]. The same argument applies to lattice Gabor frames on $L^2(\mathbb{R}^d)$, since their index group is $\mathbb{Z}^{2d}$: rank at least two gives the same growth contradiction, while rank at most one gives the component contradiction. The wavelet half of the source question is not claimed here; the usual (j, k) wavelet family lacks the stationary abelian $\mathbb{Z}^2$ correlation law used in (3).

References

- [1] O. Christensen, M. Hasannasab, and F. Philipp, *Frame properties of operator orbits*, Math. Nachr. 293 (2020), 52–66; arXiv:1804.03438.
- [2] R. Corso, *Orbits of bounded bijective operators and Gabor frames*, Ann. Mat. Pura Appl. 200 (2021), 137–148; arXiv:2004.02152.